A Novel Two-Step Framework for Mapping Fraction of Mulched Film Based on Very-High-Resolution Satellite Observation and Deep Learning

Zhihao Wei, Yaokui Cui, Sien Li, Xuhui Wang, Jinwei Dong, Lifeng Wu, Zhaoyuan Yao, Shangjin Wang, and Wenjie Fan

***Abstract*— The fraction of mulched film is of great significance for evaluating the agricultural water-saving effect and controlling environmental plastic pollution. Unfortunately, there is no work that has been done to obtain this parameter due to the mixed pixel issue of satellite imagery with medium and low resolutions, until now. In this study, we proposed a novel two-step framework for mapping fraction of mulched film based on very-high-resolution satellite observation and deep learning. The first step is extracting the extent of the mulched film based on a new few-shot learning model named parameter transition convolutional neural network (PT-CNN), which aims to increase the extraction accuracy and overcome the lack of labeled training data. The second step is retrieving the fraction of the mulched film at pixel scale based on a spectrum analysis method. The result shows that the proposed PT-CNN outperforms several state-of-the-art methods in mulched film extracting, with $F1$-scores of 97.09% and 98.65% for white and black mulched film, respectively. Meanwhile, the retrieved pixel scale fraction of mulched film shows high consistency to in situ measurement, with mean absolute error (MAE) of 0.0321. The proposed method can be useful in agricultural water resource management and environmental governance.**

***Index Terms*— Agricultural plastic mulch, deep learning, fraction of mulched film, remote sensing, very high resolution.**

I. Introduction

PLASTIC-MULCHED film has played a crucial role in agriculture in global cold and arid regions [1] and has the advantages of reducing evapotranspiration [2], [3], [4]; conserving soil moisture [5]; modifying soil temperature [6]; saving irrigation water [7], [8], crop antifreeze [9], and weed suppression [10]; and significantly increasing the crop yield [11], [12], [13], [14], [15]. Since the first mulched film was invented in the 1950s, it has gradually become one of the most widely used agricultural materials in arid regions among America, Europe, the Middle East, North Africa, and China. The mulched film is the most economical and efficient solution for arid agricultural regions around the world. Fig. 1 shows the white and black mulched film used in the center region of the Shiyanghe River Basin, China. In 2021, China had more than 17 million hectares of film mulching farmland [16], accounting for 75% of the world's total. The mulched film in China increases 30 million tons of crop yield per year, which can feed additional 80 million people. On the one hand, the film mulching technique can increase 45% of the crop yield and save 58% of the irrigation water. On the other hand, since mulched film usually takes more than 100 years to decompose in the natural environment, the recent huge usage of mulched film has resulted in serious challenges of dealing with waste mulched film, which causes environmental pollution problems and has drawn lots of attention from the government side [17], [18], [19].

The fraction of mulched film is the fraction of ground mulched by plastic film [20]. It can detailedly describe how much percentage of the farmland is covered by mulched film. Previous studies indicate that the fraction of mulched film highly influences the consumption of irrigation water [21], [22], [23], [24]. Moreover, water consumption can be overestimated if the fraction of mulched film at pixel scale is missing during the estimation [25], [26]. Thus, obtaining the fraction of mulched film at pixel scale plays an important role in agricultural water resource management, environmental protection, and human health [27], [28].

Retrieving the fraction of mulched film at pixel scale requires the extraction of the relevant pixels [29], [30]. There are some works that have been done on mulched film extraction. However, none of them has retrieved the fraction of mulched film at pixel scale, until now. This is because state-of-the-art methods are mostly working on mulched film extraction with medium- and high-resolution satellite imagery [31], [32]. The retrieval of the fraction for mulched film remains unsolved, due to the insufficient resolution of the medium-

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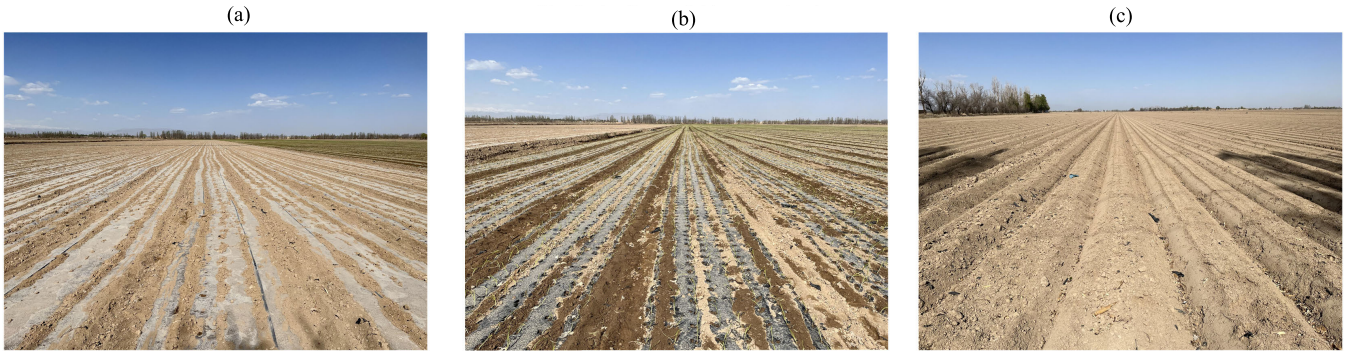


Fig. 1. Different types of mulched film in farmlands in arid regions of China. (a) White film mulching farmland. (b) Black film mulching farmland. (c) Unmulched farmland.

and high-resolution satellite imagery [33], [34]. The very-high-resolution satellite imagery could provide much more detailed spectral information for the mulched film [35], [36] and shows the possibility of retrieving the fraction of mulched film toward each farmland of a certain region, yet still requires further research on the designation of the retrieval method.

Generally, there are mainly two methods for mulched film extraction. First, many existing studies utilize threshold-based indices for mulched film extraction with remote sensing data. Fu et al. [37] proposed a modified plastic-mulched cropland index based on high-resolution imagery, including Sentinel-2 and Landsat 8 for the Shiyanghe River Basin of China. Xiong et al. [38] designed a plastic-mulched farmland index based on multiresolution satellite imagery, including Sentinel-2, Landsat 7, Landsat 8, and moderate-resolution imaging spectroradiometer (MODIS), and applied the index to extracting the mulched film of a region in northwest China. However, the performance of the threshold-based indices could be unstable during large-scale mulched film extraction, especially when dealing with mixed pixels of medium and high satellite imagery [39], [40]. Second, the traditional machine learning method has also been used for mulched film extraction. Wang and Lu [41] utilized the random forest (RF) technique to detect plastic-mulched land cover in northwest China. Hasituya and Chen [42] used RF and support vector machine (SVM) algorithms to extract mulched film based on Landsat 8 satellite imagery of two regions located in southwest and east China. Lu et al. [43] utilized the decision tree classifier to classify Landsat-5 TM Imagers and mapped the plastic-mulched land cover of a region in northwest China. Compared with threshold-based indices, the traditional machine learning method could utilize each spectral band during model calculation. However, the existing methods cannot directly apply to very-high-resolution satellite imagery for mulched film extraction due to the lack of a sufficient number of labeled samples. Moreover, there could be two scenarios for mulched film mapping, as the object-classification-based method and the pixel-fraction-based method. Due to the mixed pixel issue in the satellite imagery, the classification-based method could overestimate the usage of mulched film. Thus, it is necessary to develop a pixel-fraction-based method, which can provide a more reliable assessment of mulch usage and holds increased importance in environmental protection and water resource management.

The challenges of mulched film extraction based on very-high-resolution satellite imagery can be summarized as two points: first, how to achieve accurate extraction while the training data is insufficient, and second, how to utilize the rich spatial information to improve the accuracy of the extraction. On the one hand, traditional machine learning-based mulched film extraction methods cannot directly use this spatial information [44]. On the other hand, the recent developments in machine learning provide new structures with the concept of deep learning, such as fully convolutional neural networks (CNNs) and recurrent neural networks [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55]. These deep learning models can utilize the concept of receptive field by capturing the spatial relationship among each current pixel with their surrounding pixels and significantly improve the model accuracy [56]. However, training deep learning models usually requires thousands of samples along with accurate labels [3], which is conflicting with the few labeled samples within the mulched film extraction task. Moreover, the common remote sensing imagery datasets do not cover the mulched film topic, which cannot directly solve the lack of mulched film samples and labels.

In this work, a novel two-step framework is proposed for retrieving the fraction of mulched film at pixel scale based on very-high-resolution satellite imagery. It includes the parameter transition (PT)-CNN-based mulched film extraction method and a spectrum analysis-based fraction of mulched film pixel scale retrieval method. The results of the proposed method are further validated by the in situ measurement and compared with several state-of-the-art methods, including the traditional machine learning method and recent deep learning method.

The rest of this article is organized as follows. Section II introduces the study area and data. Section III describes the methodology. Section IV shows the result. Section V draws the discussion. Section VI concludes this article.

II. Study Area and Data

A. Study Area

Four regions in western China were selected as the study areas [Fig. 2(a)], including Wuwei city, Zhangye city, Jiuquan city in Gansu Province, and Aksu city in Xinjiang Province. The study area of Wuwei is located in the center region

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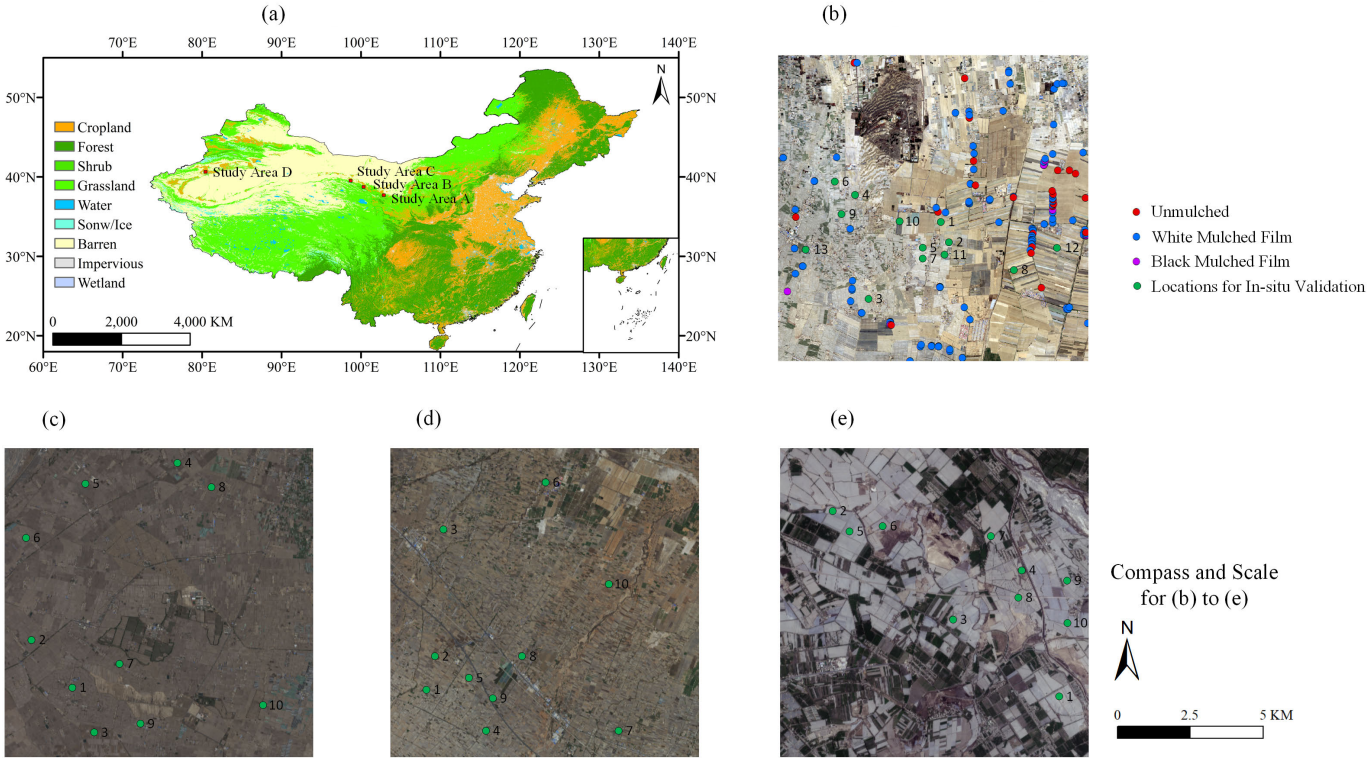


Fig. 2. Location of the study areas and in situ validation. (a) Location of four study areas in China, including (b) study area A in Wuwei, (c) study area B in Zhangye, (d) study area C in Jiuquan, and (e) study area D in Aksu. (b)–(e) BJ-2 satellite imageries and visualized by using red, green, and blue bands. The green points in (b)–(e) represent the in situ measurement location for validation.

of the Shiyanghe River Basin, covering 100-km^2 region of Wuwei, starting from 37.77°N to 37.87°N and from 102.81°E to 102.94°E [Fig. 2(b)]. The study area of Zhangye spans from 38.81°N to 38.91°N and from 100.32°E to 100.46°E [Fig. 2(c)]. The study area of Jiuquan spans from 39.56°N to 39.67°N, 98.66°E to 98.78°E [Fig. 2(d)]. The study area of Aksu spans from 40.62°N to 40.72°N and from 80.41°E to 80.53°E [Fig. 2(e)]. The geographic environment of these four study areas has similarities and differences. On the one hand, these areas have an arid temperate continental monsoon climate, characterized by less precipitation and large evaporation. This similar climate environment in all four study areas necessitates a significant usage of mulched film on farmland. On the other hand, the agricultural economy and city development are different in these study areas. This difference can be found in the agricultural mechanization level, the spatial distribution of the farmland, and finally results in unique characteristics in the regional mulched film distribution and fraction. The in situ measurement of the reflectance for the mulched film and other objects is shown in Fig. 3. These objects are collected in the study area of Wuwei, and the reflectance of the objects is measured by using spectroradiometers (ASD FieldSpec). It can be seen that the reflectance among mulched film and other objects is separable.

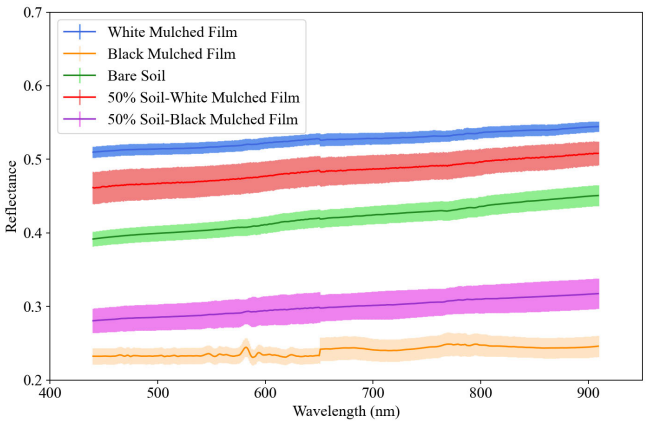


Fig. 3. In situ measurement of reflectance for the mulched film and other objects. The 50% soil-white mulched film and the 50% soil-black mulched film are the reflectance of the sampling scene with half of bare soil with white mulched film and black mulched film, respectively.

B. Ground Data

Beijing-2 satellite was scheduled to acquire the study areas' imagery, including Wuwei on April 17, 2022; Zhangye on April 28, 2023; Jiuquan on April 28, 2023; and Aksu on April 30, 2023. These dates were carefully arranged according to the weather forecast and the local sowing date, which are later than the completion of the local farmland mulching, and earlier than the local crop germination. The parameters of the Beijing-2 satellite are shown in Table I.

The Beijing-2 satellite can acquire the very-high-resolution imagery, including the panchromatic band at 0.8-m and multispectral bands at 3.2-m spatial resolution. Fig. 2(b)–(e) shows the false-color composite image of four study areas, using the red, green, and blue bands of the Beijing-2 imagery. The false-color composite images in the rest of this article

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TABLE I
PARAMETERS OF THE BEIJING-2 SATELLITE IMAGERY

Parameter	VALUE
Launch date	10th July 2015
Orbit height	651 km
Spatial resolution	Panchromatic band: 0.8 m Multispectral bands: 3.2 m
Band Range	Panchromatic: 450-650 nm, Red: 440-510 nm, Green: 510-590 nm, Blue: 600-670 nm, NIR: 760-910 nm

follow the same format. It can be seen that these scenes of Beijing-2 imagery have very high quality, with no cloud or cloud shadow. Moreover, each scene contains rich types of surface objects, such as building, farmland with different types of mulched film, sand, grassland, and road.

C. In Situ Measurement

The in situ measurement for four study areas was investigated at the same date as the Beijing-2 imagery. In order to ensure that the sampling locations are distributed throughout the entire region of each study area, the path of the field survey was carefully designed. During the field survey, the geographic coordinates and the detailed conditions of the mulched film were recorded, such as the color of the film mulch, the average width, and the spacing of the mulched film with in a single farmland. Fig. 2(b)–(e) shows the investigated locations of four study areas, which are marked by green points.

In detail, the average fraction F'_{MF_i} used as in situ measurement during the validation is defined as

$$F'_{\mathrm{MF}_i} = \frac{L_m}{L_m + L_s} \tag{1}$$

where L_m represents the average width of the mulched film and L_s represents the average spacing between each column of the mulched film.

III. METHODOLOGY

We proposed a novel two-step framework to retrieve the fraction of mulched film at pixel scale based on very-high-resolution satellite imagery. It includes a PT-CNN model for mulched film extraction and a spectrum analysis method for retrieving the fraction of mulched film at pixel scale. The flowchart of the proposed framework is shown in Fig. 4.

First, the imagery (here is the very-high-resolution satellite BJ-2 imagery) is preprocessed. Then, the mulched film within the imagery is extracted based on the proposed PT-CNN model. Specifically, in order to guarantee the robustness of the PT-CNN, the available unlabeled samples from all study areas are used in the core step of PT-CNN training. Third, the fraction of mulched film is retrieved based on the spectrum analysis of the very-high-resolution satellite imagery and the mulched film extraction result.

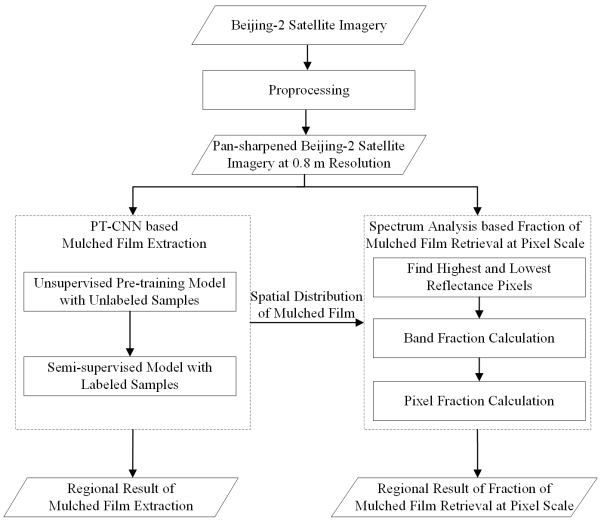


Fig. 4. Flowchart of the proposed two-step framework for fraction of mulched film pixel scale retrieval.

TABLE II
DETAILS OF THE MANUAL LABELED DATASET

Parameter	Value
Sample quantity	7056
Sample size	288×288×4 pixels
Label quantity	85
Label size	288×288×1 pixels
Spatial resolution of sample	0.8 m/pixel
Pixel values within each label	0: Unmulched 1: White agricultural plastic mulch 2: Black agricultural plastic mulch
Data source	Beijing-2 satellite

A. Preprocessing

First, the very high spatial resolution multispectral imagery is pan-sharpened to 0.8 m by the panchromatic band. Then, a very-high-resolution mulched film dataset, along with pixel-level manual annotation, is presented. Specifically, the four scenes of the Beijing-2 imagery with total pixels of 12 096 × 12 096 are cropped into image samples with 288 × 288 pixels. In addition, the accurate manual annotations of the mulched film in the study area of Wuwei are labeled via visual interpretation and in situ survey. The details of the present dataset are shown in Table II.

1) PT-CNN Model for Mulched Film Extraction: The structure of PT-CNN model for mulched film extraction is shown in Fig. 5.

The method includes two steps. First, the unsupervised pretraining model is used to pretrain the parameters within the model using the massive unlabeled samples. Second, the front layers of the semi-supervised model are set by the pretrained parameters and trained by the few labeled samples. The details of these two steps are explained as follows.

2) Building Unsupervised Pretraining Model With Unlabeled Samples: The idea of the unsupervised pretraining

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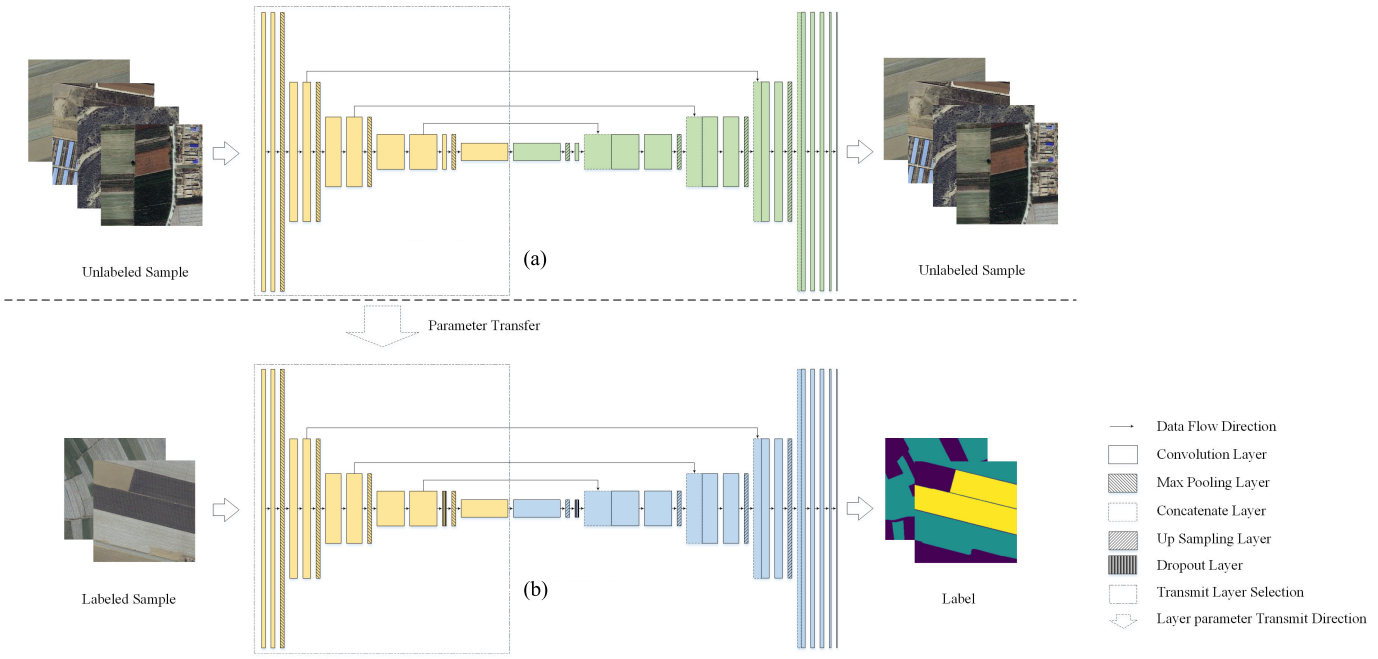


Fig. 5. Structure of the proposed PT-CNN model. The extraction model includes two steps. (a) First step is the unsupervised pretraining model training with unlabeled samples based on structure. (b) Second step is the semi-supervised model training for film mulch extraction based on the structure.

model is leveraging domain knowledge from the unlabeled samples and using it to fine-tune the semi-supervised model. The unsupervised pretraining model structure is shown in Fig. 5(a). The unlabeled samples contain different types of land covers. The unsupervised pretraining model is designed based on the U-net structure component.

The unsupervised pretraining model contains 33 layers, including convolution layers, max-pooling players, concatenate layers, upsampling layers, and dropout layers. The kernel size of the convolution layers is 3, and the pool size of the pooling layers is 2. The initializer of each layer is set as normal initialized. The activation of the last convolutional layer is using the sigmoid function. First, the domain knowledge is extracted by the convolution layer and max-pooling layer. Then, the concatenate layer and the max-pooling layer are used to reconstruct the input sample. In addition, the concatenate layer is used to connect the convolution layer and provide more information for the image reconstruction process. The dropout layer is used to prevent overfitting during the model training. The selected layers' parameters shown in Fig. 5(a) are further transmitted to the semi-supervised model in Fig. 5(b). The loss function is defined as

$$\text{loss} = n^{-1} \sum_{i=1}^{n} \left(p_i - p_i'\right)^2 \tag{2}$$

where the training loss is the mean square error of the label p_i toward model prediction p_i' within a training batch of size n.

3) Building Semi-Supervised Model for Mulched Film Extraction: The semi-supervised model is trained by the labeled samples, as shown in Fig. 5(b). There are 33 layers in semi-supervised model. Different from the unsupervised pretraining model, the activation of the last convolutional layer in semi-supervised model is using the softmax function. The initializer of each layer is set as normal initialized. Before the model training, the semi-supervised model is first set by the pretrain parameters, which are transmitted from the unsupervised pretraining model. Then, the layers with pretrained parameters are used to extract the domain knowledge of the labeled samples. Finally, the domain knowledge is used for model training. The loss function is defined as

$$\text{loss} = -\sum_{i=1}^{n} p_i \cdot \log p_i' \tag{3}$$

where the training loss is the categorical cross entropy of the label p_i toward model prediction p_i' within a training batch of size n.

B. Retrieving Fraction of Mulched Film at Pixel Scale Based on Spectrum Analysis

Assuming that the reflectance within the mulched film pixel in farmland is defined as ρ_1, the reflectance toward the bare soil pixel is defined as ρ_2, the fraction of the mulched film toward current pixel i is defined as f_{MF_i}, and the fraction of the bare soil within current pixel is calculated as $1 - f_{\text{MF}_i}$. Thus, the current pixel's reflectance ρ_i can be calculated as

$$\rho_i = f_{\text{MF}_i} \times \rho_1 + \left(1 - f_{\text{MF}_i}\right) \times \rho_2. \tag{4}$$

Then, the fraction of the film mulching f_{MF_i} toward pixel i can be derived from (4) into the following equation:

$$f_{\text{MF}_i} = \frac{\rho_i - \rho_2}{\rho_1 - \rho_2}. \tag{5}$$

Specifically, among all the farmland pixels that contain mulched film and bare soil reflectance information, we assume that the pixel with pure white mulched film has the highest

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reflectance, and the pixel with pure bare soil has the lowest reflectance. Thus, (5) can be further transformed into the following equation:

$$f_{\mathrm{MF}_i} = \frac{\rho_i - \rho_{\min}}{\rho_{\max} - \rho_{\min}} \tag{6}$$

where $\rho_{\max}$ and $\rho_{\min}$ are present the pixels with the highest reflectance and the lowest reflectance within the farmland satellite imagery sample, respectively.

Furthermore, since the satellite imagery contains multiple spectral bands, (6) is further transformed into the following equation:

$$f_{\mathrm{MF}_{b,i}} = \frac{\rho_{b,i} - \rho_{b,\min}}{\rho_{b,\max} - \rho_{b,\min}} \tag{7}$$

where three visible spectral bands, red, green, and blue are used during the retrieval, which are the spectral bands that are commonly used in mulched film extraction [57]. The fraction $f_{\mathrm{MF}_{b,i}}$ for pixel within each spectral band b is calculated based on (7). Here, $\rho_{b,i}$ represents the reflectance of pixel i within the spectral band b; $\rho_{b,\max}$ and $\rho_{b,\min}$ represent the maximum and minimum reflectance among the pixels within the spectral band b, respectively.

Then, the fraction for white mulched film F_{MF_i} for each pixel i is calculated by the following equation, which is the average of the fraction $f_{\mathrm{MF}_{b,i}}$ among each optical band b:

$$F_{\mathrm{MF}_i} = \frac{1}{n}\sum_{b=1}^{b=n} f_{\mathrm{MF}_{b,i}} \tag{8}$$

where n represents the total quantity of the optical spectral bands of the satellite imagery sample. Finally, the average fraction of white mulched film F_{MF} is calculated by the following equation, which is the average fraction of white mulched film for F_{MF_i} among all the pixels within the sample:

$$F_{\mathrm{MF}} = \frac{1}{m}\sum_{i=1}^{i=m} F_{\mathrm{MF}_i} \tag{9}$$

where m represents the total quantity of the pixels within the average white mulched film farmland of the imagery sample. In addition, the retrieval of the fraction of black mulched film follows a similar process to that shown above. However, since the reflectance of the pure black mulched film is lower than the pure bare soil, the final fraction of black mulched film can be further given by the following equation:

$$F_{\mathrm{MF}_{\mathrm{black}}} = 1 - F_{\mathrm{MF}}. \tag{10}$$

C. Accuracy Assessment

The $F1$-score, precision, and recall are used to evaluate the mulched film extraction results. The calculation of these variables is as follows:

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{11}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{12}$$

$$F_1\mathrm{score} = \frac{2 * \mathrm{Precision} * \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{13}$$

where TP, FP, and FN represent the true positive, false positive, and false negative results of each pixel of the samples during mulched film extraction, respectively.

Moreover, the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE) are used to further validate the average fraction of mulched film F_{MF} as follows:

$$R = \left(\frac{\sum_{i=1}^{n}\left(F_{\mathrm{MF}_i}{}' - \overline{F_{\mathrm{MF}_i}{}'}\right)\left(F_{\mathrm{MF}_i} - \overline{F_{\mathrm{MF}_i}}\right)}{\sqrt{\sum_{i=1}^{n}\left(F_{\mathrm{MF}_i}{}' - \overline{F_{\mathrm{MF}_i}{}'}\right)^2} \times \sqrt{\sum_{i=1}^{n}\left(F_{\mathrm{MF}_i} - \overline{F_{\mathrm{MF}_i}}\right)^2}} \right)^2 \tag{14}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(F_{\mathrm{MF}_i} - F_{\mathrm{MF}_i}{}'\right)^2} \tag{15}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|F_{\mathrm{MF}_i}{}' - F_{\mathrm{MF}_i}\right| \tag{16}$$

where F'_{MF_i} is the ground truth and $\overline{F_{\mathrm{MF}_i}{}'}$ is its mean, and F_{MF_i} is calculated fraction of mulched film and $\overline{F_{\mathrm{MF}_i}}$ is its mean.

IV. Result

The limited quantity of labeled training samples and the unsolved method of fraction retrieval are two important challenges in retrieving the very-high-resolution mulched film pixel scale fraction. First, the result of the proposed PT-CNN model is compared with several state-of-the-art methods on the mulched film extraction task. Then, the fraction of mulched film pixel scale retrieval method is validated by in situ measurement. Finally, the regional results of the mulched film extraction and fraction retrieval are generated and analyzed.

A. Overall Performance of PT-CNN

1) Performance of Unsupervised Pretraining Model: Due to the image restoration ability of the pretrained model being crucial to the PT-CNN, the performance of the pretrained model is shown first. In this study, a total number of 2400 unlabeled samples from four study areas are randomly selected and used for unsupervised pretraining model training and testing. First, the randomly selected 2000 samples are used for unsupervised pretraining model training. Then, the remaining 400 samples are used for the unsupervised pretraining model testing.

The training loss curve of the unsupervised pretraining model is shown in Fig. 6(b). It can be seen that the mean square error loss reduces during each epoch of training and finally becomes stable after 80 epochs.

Furthermore, the input and the output of the unsupervised pretraining model during the model testing are shown in Fig. 7. The first row is the unlabeled sample input of the model, and the second row is the output of the model.

Each column represents the corresponding unlabeled sample input and output pair of the model. These unlabeled sample inputs contain scenes including building, mountain field, film mulching farmland, soil, and tree. It can be seen that the

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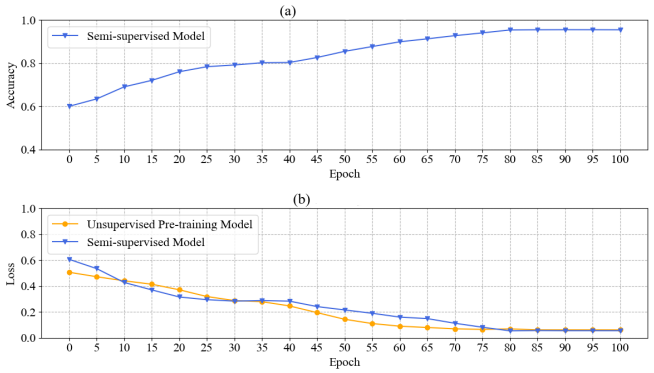


Fig. 6. Training accuracy and loss curves of the PT-CNN model, including the unsupervised pretraining model and semi-supervised model. (a) Training accuracy. (b) Training loss.

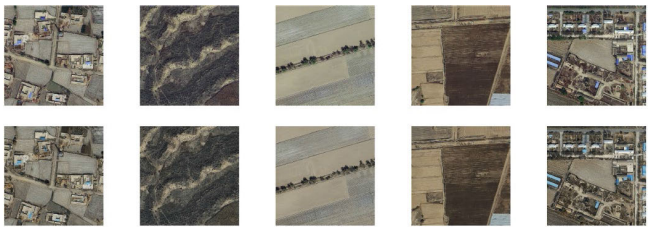


Fig. 7. Visualization of the image reconstruction performance of the unsupervised pretraining model. The first row is the unlabeled sample input of the unsupervised pretraining model, and the second row is the output of the unsupervised pretraining model.

TABLE III
DETAILS OF THE MANUAL LABELED DATASET

Method	Parameter	Value
RPGI	Threshold	NDVI <1.73, RPGI > 1.6
SVM	Kernel	Radial Basis Function (RBF)
RF	Criterion	Gini
Pspnet	Learning Rate	0.0001
	Loss Function	Categorical Cross Entropy
	Batch Size	4
Resnet	Learning Rate	0.0001
	Loss Function	Categorical Cross Entropy
	Batch Size	4
Unet	Learning Rate	0.0001
	Loss Function	Categorical Cross Entropy
	Batch Size	4
MRUnet	Learning Rate	0.0001
	Loss Function	Categorical Cross Entropy
	Batch Size	4
PT-CNN	Learning Rate	0.0001
	Loss Function	Mean Square Error, Categorical Cross Entropy
	Batch Size	4

false-color composite image of the input and the output shows high similarity, which proves the image reconstruction performance of the unsupervised learning model based on the extracted knowledge from the unlabeled samples.

2) Validation of the PT-CNN Against In Situ Measurement: A total number of 85 labeled samples are divided into three groups, including 40 labeled samples for semi-supervised model training, 22 labeled samples for model evaluation, and 23 labeled samples for model testing, with a ratio of training, evaluation, and testing as 5:2:3.

The training accuracy curve and loss curve of the PT-CNN model are shown in Fig. 6(a) and (b), respectively. It can be seen that the accuracy increases during the first 80 epochs and finally reaches a stable level after the 80th epoch. The loss curve decreases following the same trend, while the accuracy curve increases. This proves that the semi-supervised model is well trained.

The PT-CNN model is compared with several state-of-the-art methods, including the spectral indices retrogressive plastic greenhouse index (RPGI) [39]; the SVM [58]; RF [59]; and several widely used deep learning models, including PSPnet (PSP) [60], Resnet [61], Unet (UNET) [62], and MultiResUnet (MRUnet) [63]. The parameters of the PT-CNN and comparison methods are shown in Table III.

The RPGI index is a binary index and is used for plastic cover detection within farmland. The calculation of the RPGI includes three steps. First, the normalized difference vegetation index (NDVI) is calculated for each pixel within the sample. Then, the RPGI is calculated for the pixels where the NDVI is lower than 1.73. Finally, the pixels with an RPGI larger than 1.6 are recognized as plastic cover [39]. Since the RPGI is a binary index, the white and black mulched film cannot be discriminated by the RPGI. Moreover, the RPGI is designed with a lower threshold, which can select the white plastic cover with high reflectance, yet misses black plastic cover with low reflectance. Thus, during the result analysis, the result of the RPGI only counts for the white mulched film.

The comparison results between the proposed method and the state-of-the-art methods are shown in Table IV. It can be seen that the deep learning methods perform better than the traditional machine learning methods SVM and RF. Furthermore, by using the domain knowledge extracted from the unlabeled samples, the PT-CNN can better recognize the mulched film compared with deep learning methods.

In the meantime, several samples are randomly selected from the group testing and evaluation group, and the mulched film extraction results for each sample based on the PT-CNN and comparison methods are shown in Fig. 8. It can be seen that the SVM and RF methods are confused by the complex texture within the spectrum bands among both the film mulching region and the unmulched region, resulting in lower performance for the mulched film extraction. The deep learning methods perform better than the traditional machine learning method, yet still fails to recognize several film mulching regions. The PT-CNN is able to better recognize the mulched film and obtain the complete and continuous mulched film extraction result.

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TABLE IV
PARAMETERS OF THE PROPOSED AND COMPARISON METHODS

Group	AC	CL	RPGI	SVM	RF	PSPNET	RESNET	MRUNET	UNET	PROPOSED
Evaluation	Precision	White	63.35	58.42	50.23	77.69	73.95	71.10	73.34	**81.54**
		Black	--	83.93	87.10	89.45	93.73	94.64	92.36	**97.17**
	Recall	White	33.81	76.89	85.09	91.22	86.33	91.91	91.28	**94.56**
		Black	--	95.87	95.30	73.53	97.05	95.58	92.14	**99.26**
	F1-score	White	38.20	60.23	55.31	62.25	75.74	72.80	75.85	**83.71**
		Black	--	87.64	89.79	76.14	95.03	94.67	91.91	**98.16**
Testing	Precision	White	87.29	89.96	85.14	94.01	95.95	95.89	95.86	**96.12**
		Black	--	81.41	85.83	96.57	97.55	98.74	86.38	**98.94**
	Recall	White	26.59	80.39	90.40	63.80	97.34	98.14	94.27	**98.21**
		Black	--	97.35	95.03	75.57	96.87	95.06	95.06	**98.37**
	F1-score	White	38.89	84.33	86.54	91,81	96.58	96.94	94.57	**97.09**
		Black	--	86.17	88.84	74.10	97.18	96.86	89.41	**98.65**

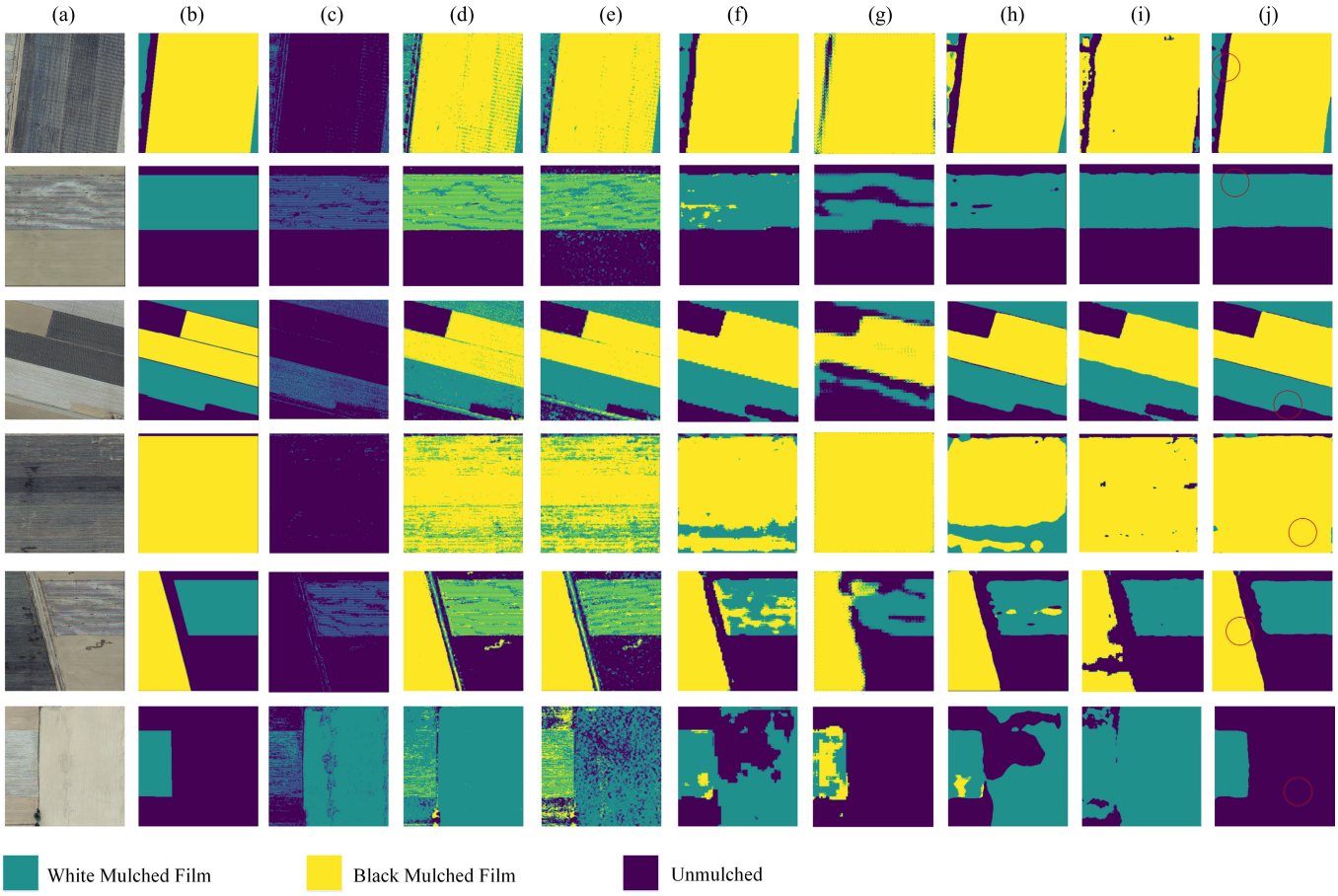


Fig. 8. Visualization of the performance of the proposed method and other state-of-the-art methods. (a) False-color composite image of the sample. (b) Visualization of the sample. (c) RPGI result. (d) SVM result. (e) RF result. (f) Resnet result. (g) PSP result. (h) UNET result. (i) MRUnet result. (j) PT-CNN result. The red circles in column (j) represent the significant improvement of the proposed method compared with other methods.

Fig. 9 shows the visualization of the $F1$-score of all the testing and evaluation samples, by comparing the PT-CNN to each comparison method. It can be seen that most of the dots are located on the right side of the scatter diagram, which indicates the better mulched film extraction performance of the PT-CNN, compared with other methods.

Furthermore, the in situ measurements are used to validate the performance of PT-CNN-based mulched film extraction. The locations of the in situ validation are shown in Fig. 2(b)–(e). In the meantime, a total of 280 additional randomly selected unmulching pixels through manual confirmation from all study areas are used for validation. The extraction result is evaluated by in situ measurement and manual visualization, which is shown in Table V. It can be seen that most of the evaluated locations are being correctly classified as white mulched film, black mulched film, and unmulched film. Moreover, the wrong classified pixels are mostly at the edge of the film mulching farmland. On the whole, it proves that the extraction result is highly consistent with the in situ measurement, which proves the performance of

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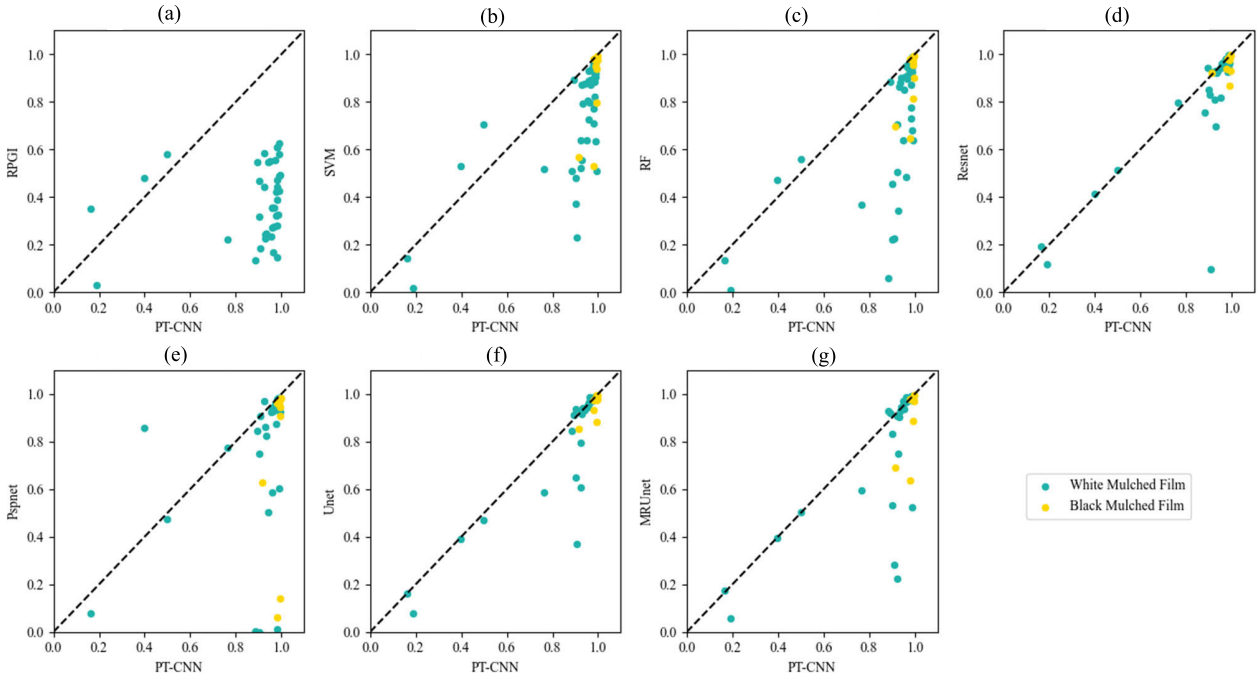


Fig. 9. Visualization of $F1$-score for the PT-CNN and other comparison methods among the test and evaluation samples. (a) PT-CNN versus RPGI. (b) PT-CNN versus SVM. (c) PT-CNN versus RF. (d) PT-CNN versus Resnet. (e) PT-CNN versus PSP. (f) PT-CNN versus UNET. (g) PT-CNN versus MRUnet.

the proposed PT-CNN during the in situ validation of mulched film extraction.

B. Validation of Retrieved Fraction of Mulched Film at Pixel Scale

The fraction for mulched film is retrieved based on the proposed method. First, Fig. 10(a) shows the false-color composite image of the sample. The in situ measurement record shows that the average width of the mulched film L_m is 80 cm, and the average spacing between each column of the mulched film L_s is 50 cm. Thus, F_{MF} for this region is 0.6153, calculated by (1). The mulched film extraction result generated by PT-CNN is shown in Fig. 10(b). Fig. 10(d) shows the histogram of the fraction for mulched film. The average of the calculated pixel scale fraction of mulched film is 0.5875, which is highly similar to the average of the in situ fraction of mulched film.

Furthermore, the in situ measurement is used to validate the average fraction for mulched film among different locations of four study areas. The locations of the in situ validation are shown in Fig. 2(b)–(e). As shown in Fig. 11, a total of 86 in situ measurements are compared with the corresponding retrieved average fraction for mulched film. The R^2, RMSE, and MAE are calculated based on the in situ validation. It can be seen that the retrieved fraction of mulched film has a high consistency to the in situ measurement, with R^2 of 0.93, RMSE of 0.036, and MAE of 0.0321. This proves the accuracy of the proposed fraction of mulched film retrieval method.

Finally, the retrieved pixel scale fraction of mulched film in the four study areas is shown in Fig. 12. First, it can be seen

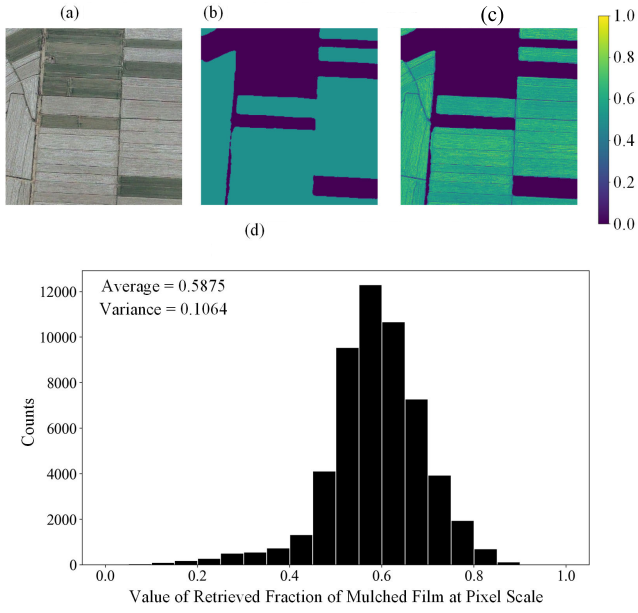


Fig. 10. Visualization of the pixel scale fraction of mulched film retrieval. (a) Sample. (b) Mulched film mask. (c) Retrieved fraction of mulched film at pixel scale. (d) Histogram of the retrieved fraction of mulched film at pixel scale.

that the mulched film within a single study area is laid within a range of fractions. The values of the pixel scale fraction of mulched film in Wuwei are mostly ranging from 0.4 to 0.8. The fraction of mulched film in Zhangye is ranging from 0.3 to 0.5. The fraction of mulched film in Jiuquan ranges from 0.3 to 0.6. In Aksu, the fraction of mulched film is ranging from 0.6 to 0.8.

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TABLE V
VALIDATION OF PT-CNN-BASED MULCHED FILM EXTRACTION BASED ON IN SITU MEASUREMENT AND MANUAL VISUALIZATION

Location	Type	White (in-situ)	Black (in-situ)	Unmulching (in-situ)
Wuwei	White (PT-CNN)	33	0	0
	Black (PT-CNN)	0	22	0
	Unmulching (PT-CNN)	2	1	31
Zhangye	White (PT-CNN)	31	0	0
	Black (PT-CNN)	0	0	0
	Unmulching (PT-CNN)	3	0	32
Jiuquan	White (PT-CNN)	21	0	0
	Black (PT-CNN)	0	25	0
	Unmulching (PT-CNN)	2	1	31
Aksu	White (PT-CNN)	39	0	0
	Black (PT-CNN)	0	0	0
	Unmulching (PT-CNN)	2	0	34

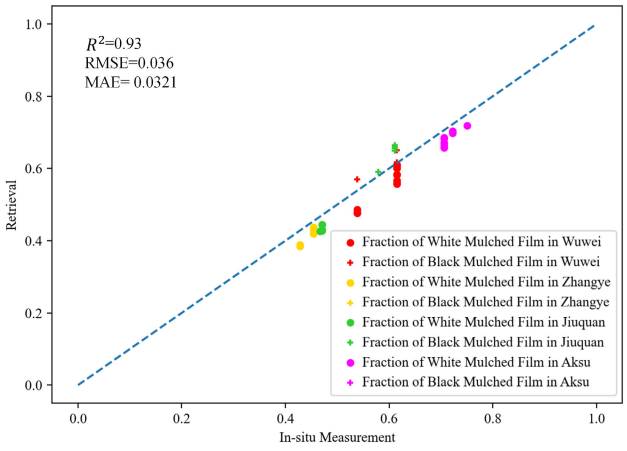


Fig. 11. Validation of the pixel scale fraction of mulched film retrieval.

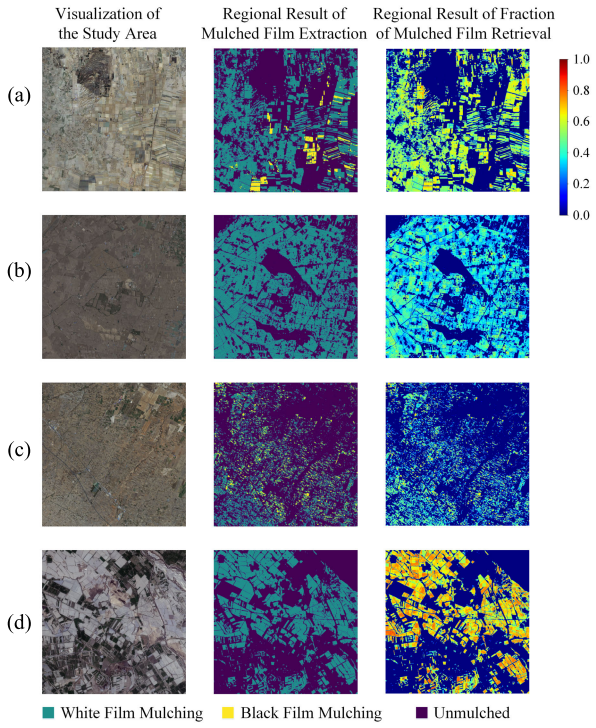


Fig. 12. Regional result of four study areas in the western part of China, located in (a) Wuwei, (b) Zhangye, (c) Jiuquan, and (d) Aksu. Each row represents the result of the corresponding area and contains three subfigures, including the false-color composite image of the Beijing-2 satellite imagery for the corresponding study area, the regional result of mulched film extraction, and the regional result of pixel scale fraction of mulched film retrieval.

Second, there is a significant regional difference in the fraction of mulched film among the four study areas. Compared with the other three study areas, Aksu has the highest average fraction of mulched film. This matches the fact that Aksu is one of the field crop planting areas in China. The cotton production mechanization is highly developed in Xinjiang Province [64], and large-width mulched films are laid by the mulching machine. Thus, the fraction of mulched film in Aksu is much higher than in other regions. Moreover, it can be seen that the distribution of mulched film farmland in Jiuquan is more scattered, compared with other study regions.

Above all, the results show that the proposed framework can well extract the mulched film and accurately retrieve its fraction among different study areas.

V. DISCUSSION

A. Importance of the Negative Samples for PT-CNN

Due to the study area containing various types of surface land cover, model training without using the negative sample that does not contain the mulched film could result in wrong unmulched region recognition toward the study area. The negative samples within the training group are shown in Fig. 13. These samples are selected from the Sand Park region of Wuwei, which is marked by the red circle in Fig. 14(a). Specifically, these samples contain the surface type that was not included in the original training group of the PT-CNN and led to the wrong classification of the unmulched region in Fig. 14(b). By adding these negative samples into the training groups for PT-CNN training, the updated mulched film extraction result is shown in Fig. 14(c). It can be seen

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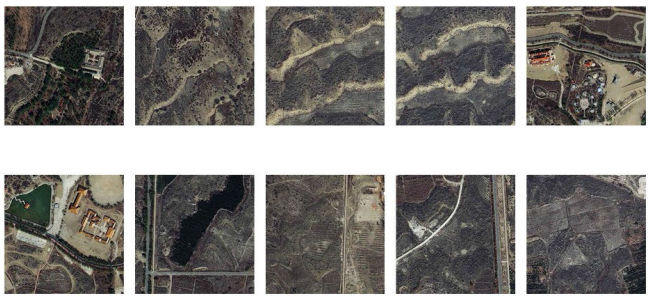


Fig. 13. Visualization of the negative samples.

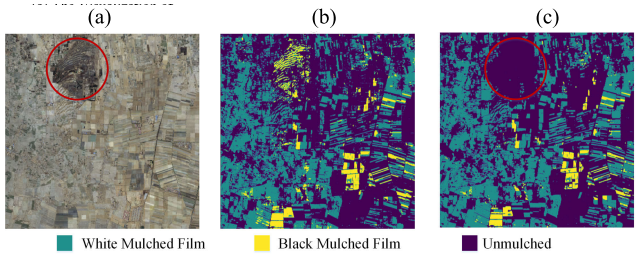


Fig. 14. Mulched film extraction result of the study area Wuwei, China. (a) False-color composite image of the Beijing-2 satellite imagery of the study area Wuwei. (b) Mulched film extraction of the study area Wuwei without using the negative samples. (c) Mulched film extraction of the study area Wuwei by adding the negative samples into the model training, which results in the correct classification of the unmulched surface region in Sand Park in the study area Wuwei, marked by a red circle.

that the entire Sand Park region has been correctly recognized as an unmulched region.

B. Advantage and Disadvantage of the Proposed PT-CNN Based Mulched Film Extraction Method

The PT-CNN model can achieve better mulched film recognition toward the very-high-resolution Beijing-2 satellite imagery. Compared with the traditional deep learning model that requires more labeled training data, the PT-CNN model is able to extract the domain knowledge by the unsupervised pretraining structure and transform this information to the semi-supervised model for better parameters modification.

The limitation of the proposed pipeline is mainly in two aspects. First, the diversity of the land surface type within the label is important toward the performance of the mulched film extraction model [65].

It can be foreseen that the PT-CNN model could be further improved for other regions by adding the labeled samples that are not contained in the study area. Second, since the time window for the mulched film laying is short, satellite imagery on a date that is shortly after the completion of the local farmland mulching, and earlier than the local crop germination is recommended to be used for mulched film extraction.

C. Advantage and Disadvantage of the Proposed Pixel Scale Fraction of Mulched Film Retrieval Method

The proposed spectrum analysis-based fraction of mulched film pixel scale retrieval method has the advantage of directly using the satellite spectrum for fraction of mulched film pixel scale retrieval, which does not need the assistance of the ground observation data. Moreover, the method is flexible and can be used on very-high-resolution satellite imagery.

However, since the width of the mulched film is usually lower than 1 m, the proposed spectrum analysis-based fraction of mulched film retrieval method could only apply to very-high-resolution satellite imagery. For insufficient spatial resolution satellite imagery, when the pixel's spatial resolution is wider than the width of the mulched film, none of the pure mulched film pixels can be found. One of the solutions could be combining the very-high-resolution and other lower spatial resolution imagery to retrieve the pixel scale fraction of mulched film. Moreover, the proposed fraction of mulched film retrieval method is currently focused on calculating the ratio between the mulched film and bare soil during the spring sowing season. In other scenes such as during the winter wheat sowing season, there could be additional snow and vegetation on farmland, which requires further research of the method.

VI. Conclusion

In this study, a novel two-step framework is proposed for retrieving the fraction of mulched film at pixel scale based on very-high-resolution satellite imagery. Specifically, the mulched film is extracted by very-high-resolution satellite imagery, and the fraction of the mulched film at pixel scale is first retrieved by our method. The method is validated in four different study areas in the western part of China.

The main conclusions are given as follows.

1) A PT-CNN model for mulched film extraction is proposed. The result shows that the PT-CNN model outperforms several state-of-the-art methods, with $F1$-scores of 97.09% and 98.65% for white and black mulched film extraction, respectively. Moreover, the in situ evaluation and manual visualization proves that the extraction result is highly consistent with the in situ measurement.
2) A spectrum analysis-based fraction of mulched film pixel scale retrieval method is proposed. The in situ measurement validation shows the good retrieval performance of the proposed method, with MAE of 0.0321.

Overall, the proposed two-step framework for mapping fraction of mulched film should be valuable in providing more detailed and effective information on agricultural water resources management and benefit the agriculture and environment. Moreover, with recent development of very-high-resolution satellite techniques, the proposed mulched film extraction and fraction retrieval method can be applied to other sources of satellite imagery, along with different conditions and types of mulched film, which can be researched in the future.

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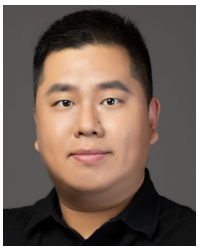
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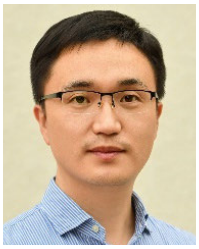
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